Container and Distributed

Elastic Load Balance (ELB)

Elastic Load Balance (ELB)

- Distributes requests **across different backend servers**.
- Supports **built-in HA.**
- Works with **Auto Scaling** to process a massive number of concurrent requests.
- Routes traffic across backend server groups based on the forwarding policies you have configured.
- Checks the **health of backend servers** to ensure that requests are routed to healthy servers.
- Supports load balancing at both **Layer 4** and **Layer 7**.

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Elastic Load Balance (ELB)

- Routes requests based on the protocol and port
you have configured.
- Matches the requests based on the forwarding rules you have configured.
    - Domain name and URL (regular expression match)
    - Advanced forwarding rules such as HTTP request method, HTTP header, and more
- Forwards the requests or returns a fixed
response to the clients based on the action you have configured.
    - Forward to a backend server group.
    - Redirect to another listener.
    - Redirect to another URL

- One or more backend servers. You can call an API to add or remove a backend server.
- **Multiple load balancing algorithms**
    - Weighted round robin
    - Weighted least connections
- Advanced settings
    - Sticky session
    - Health check

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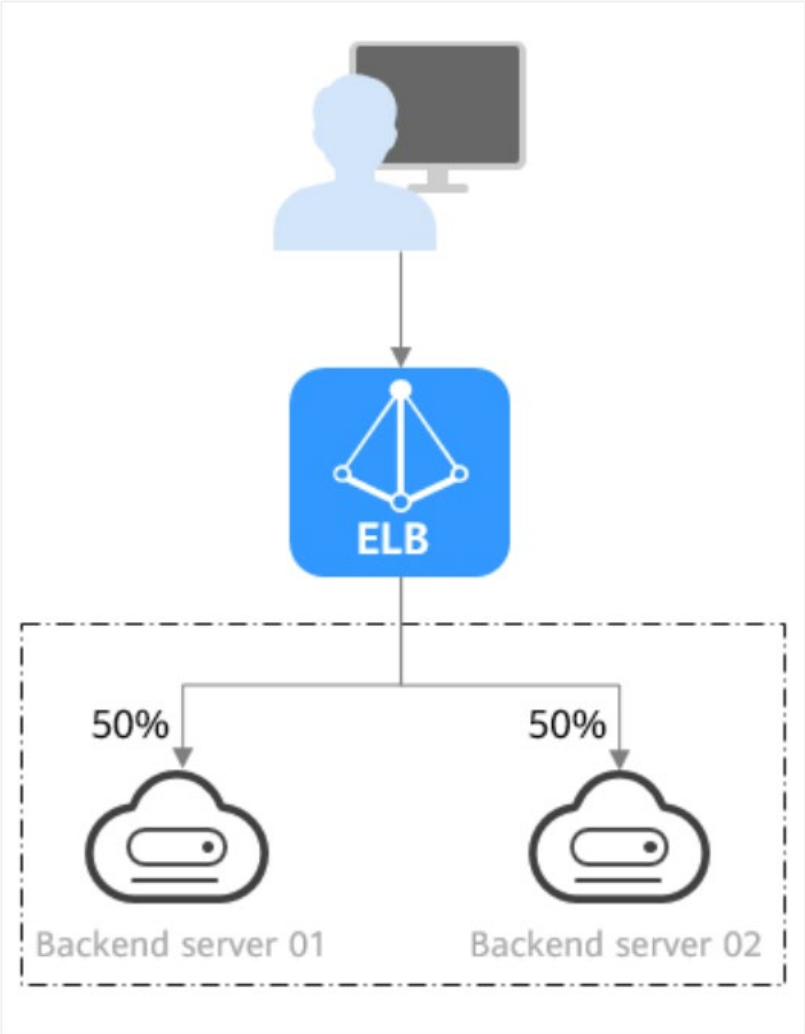
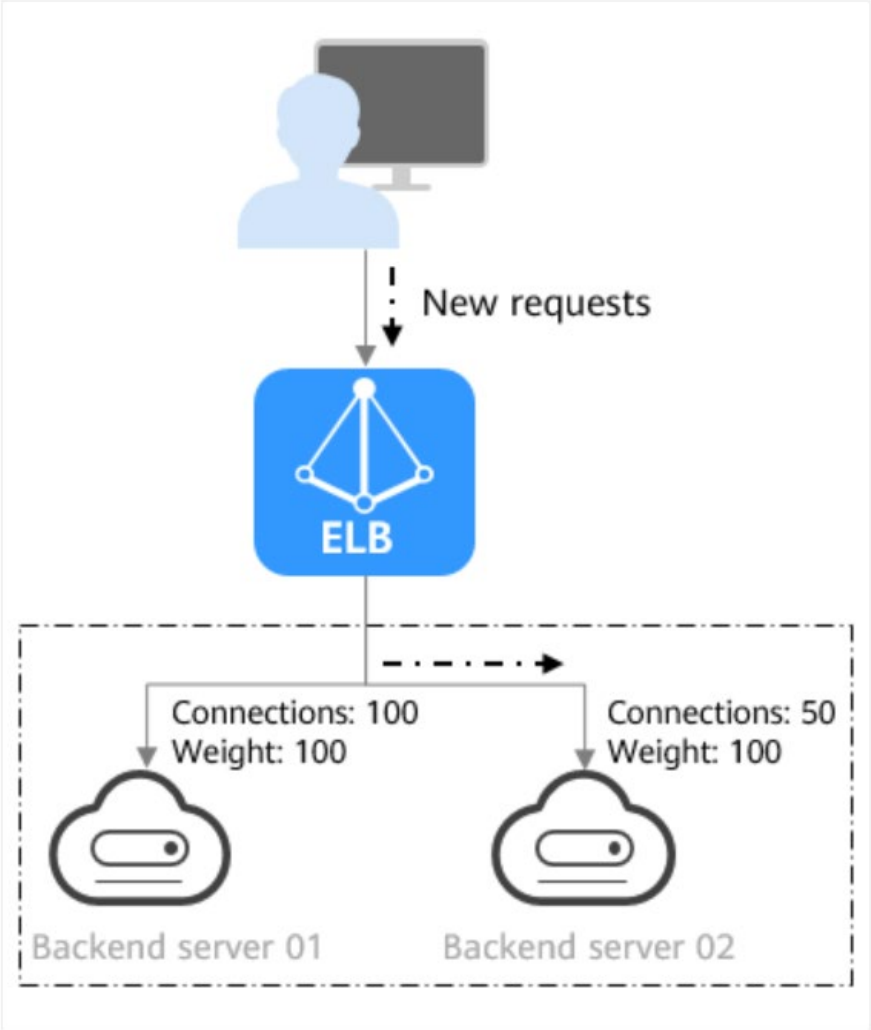
Weighted Round Robin



Weighted Least Connections



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Cloud Eye

Cloud Eye

- Operations without data courts disaster.
- Not a traditional proactive monitoring system
  - Cloud service data is reported to Cloud Eye.
  - Cloud Eye is decoupled from other services.
- Tracks health metrics for all of your services.
- Keeps you up to date with built-in dashboards and alarms

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Auto Scaling

Auto Scaling

- Can create or delete ECS instances across AZs.
- Supports a broad range of conditions and policies to create or delete instances.
- Automatically identifies and replaces unhealthy instances.
- Can work with ELB to distribute incoming traffic across
- healthy backend servers.

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Scaling Policies and Scenarios

Dynamic scaling based on performance

- Real-time monitoring of performance
- Automatic, dynamic scaling
- Reactive scaling
- Example
  Usage of CPU, Memory , etc.

Scheduled scaling based on predictable workload changes

- Regular workload changes
- Scaling by schedule
- Proactive scaling
- Example
  Schedule, Periodic

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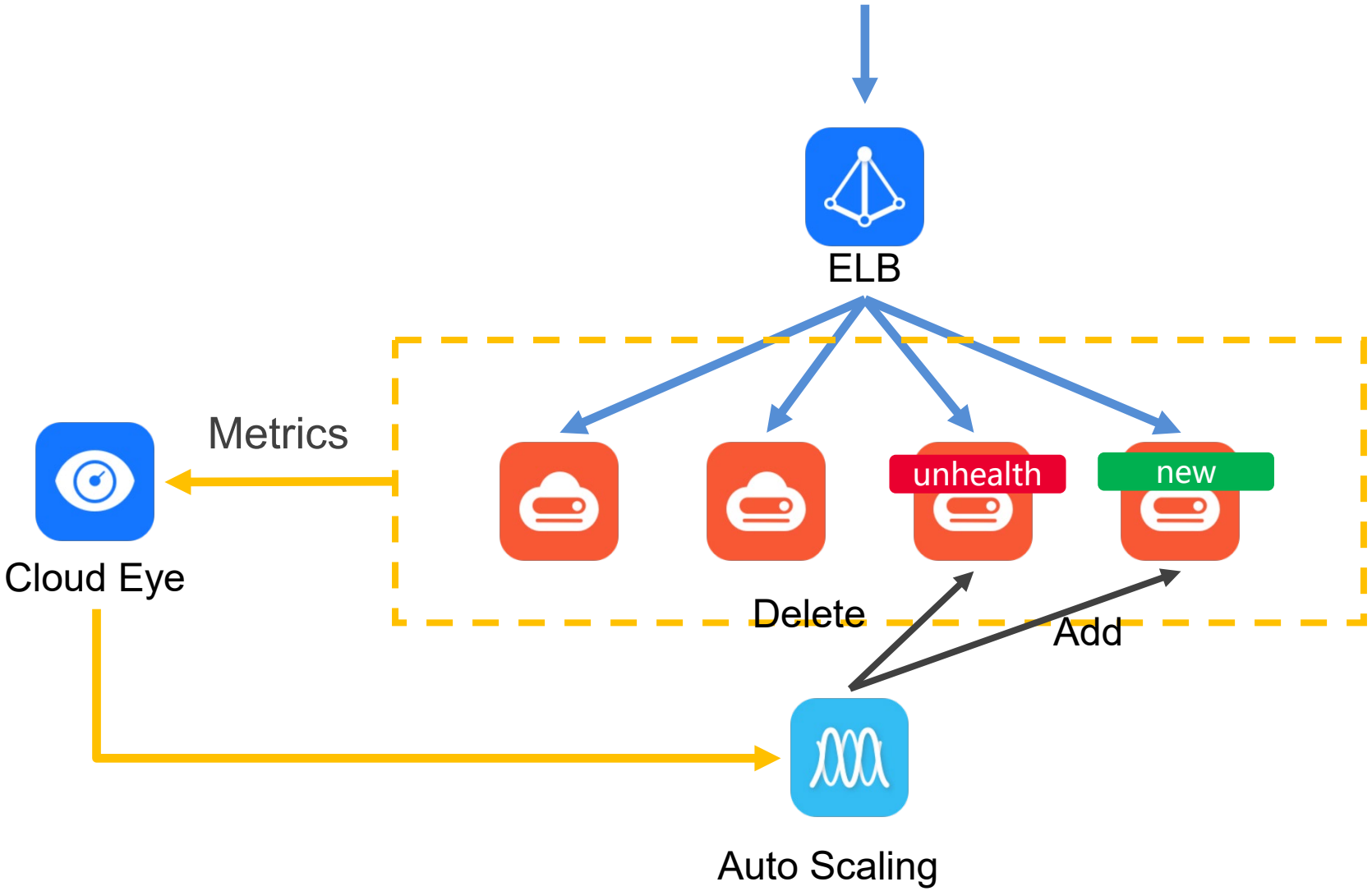
Automatically Add a healthy backend server to ensure stable performance



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Cloud Native 2.0

Cloud native is a kind of approach to design, construct and operate workloads natively in the cloud**, and it** can take full advantage of cloud computing.

Data governance
Big data, Data Lake

Compute
Container Serverless, Function

Agile DevOps

AI Platform Components

- Secure, trustworthy, efficient, and intelligent development
- Flexible **governance of dual-stack microservices**
- Co-existence of legacy and new applications; **efficient application integration**
- Serverless development allows **developers to focus on features**
- Containers help **implement service logic** and **reduce attention an coupling to servers**
- Intelligent management and efficient O&M
- Decoupled compute and storage; lower TCO
- **One-stop AI platform**; inclusive AI
- Consistent security standards for the platform and tenants

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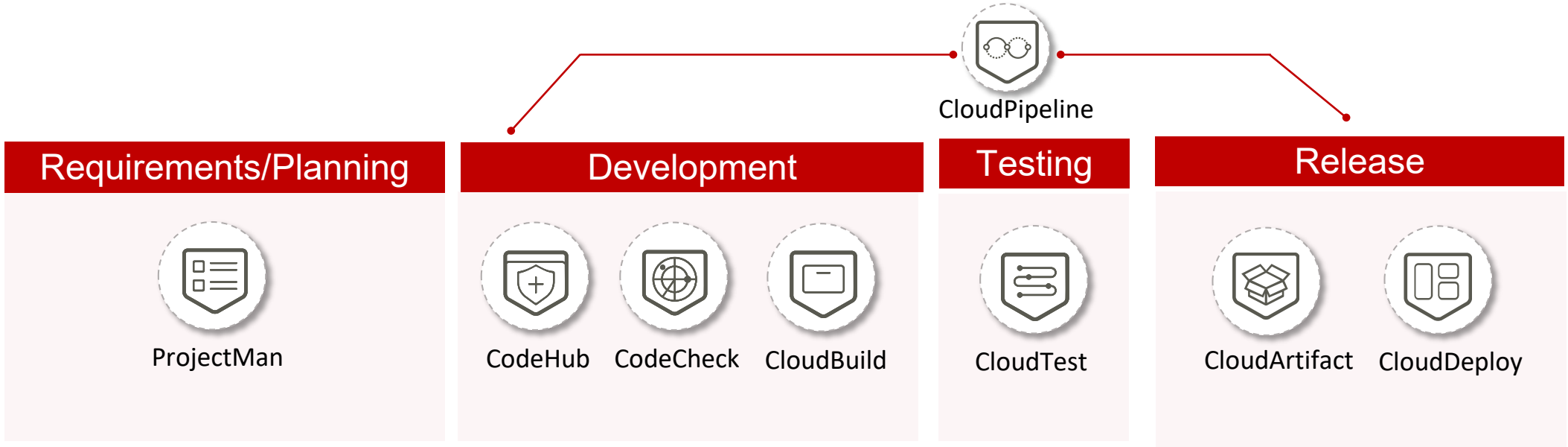
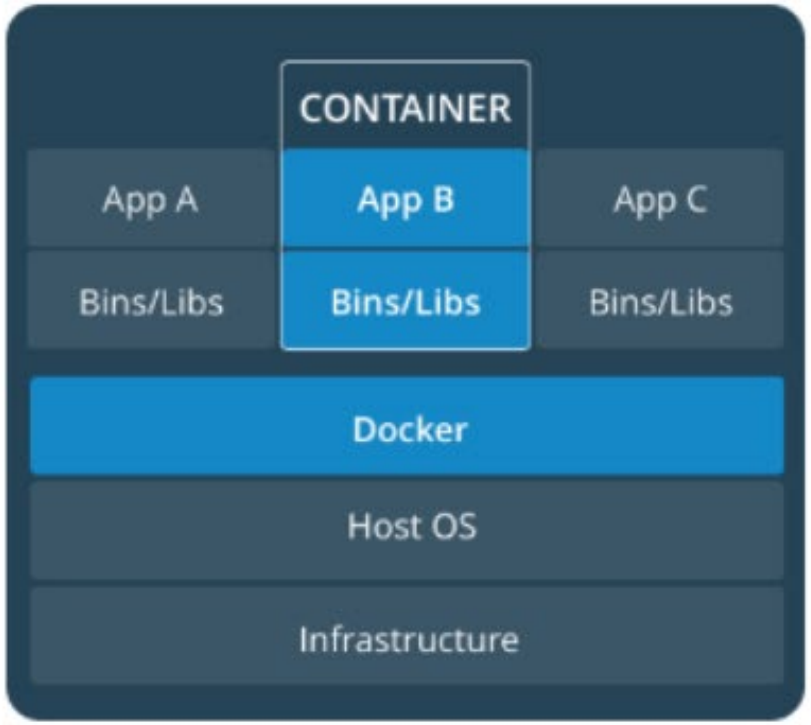
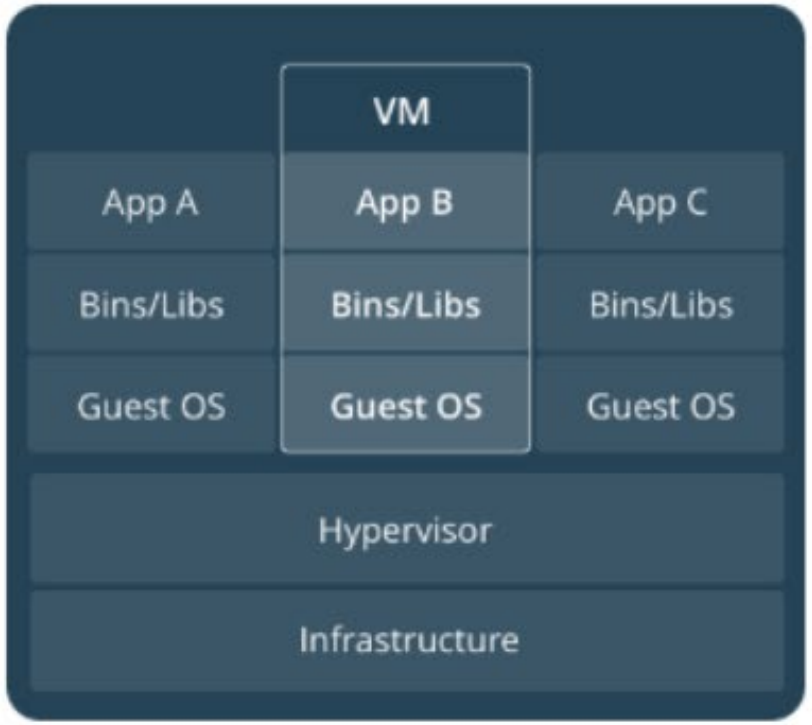
DevCloud (CodeArts)



- Huawei Cloud provides a complete **agile development system** with all required components.
- Its integrated development environment is easy to use.
- An agile development process means **quick go-to-market and quick response** to customer needs.
- Innovation needs cloud, and agility is the catalyst for innovation

What is Container?





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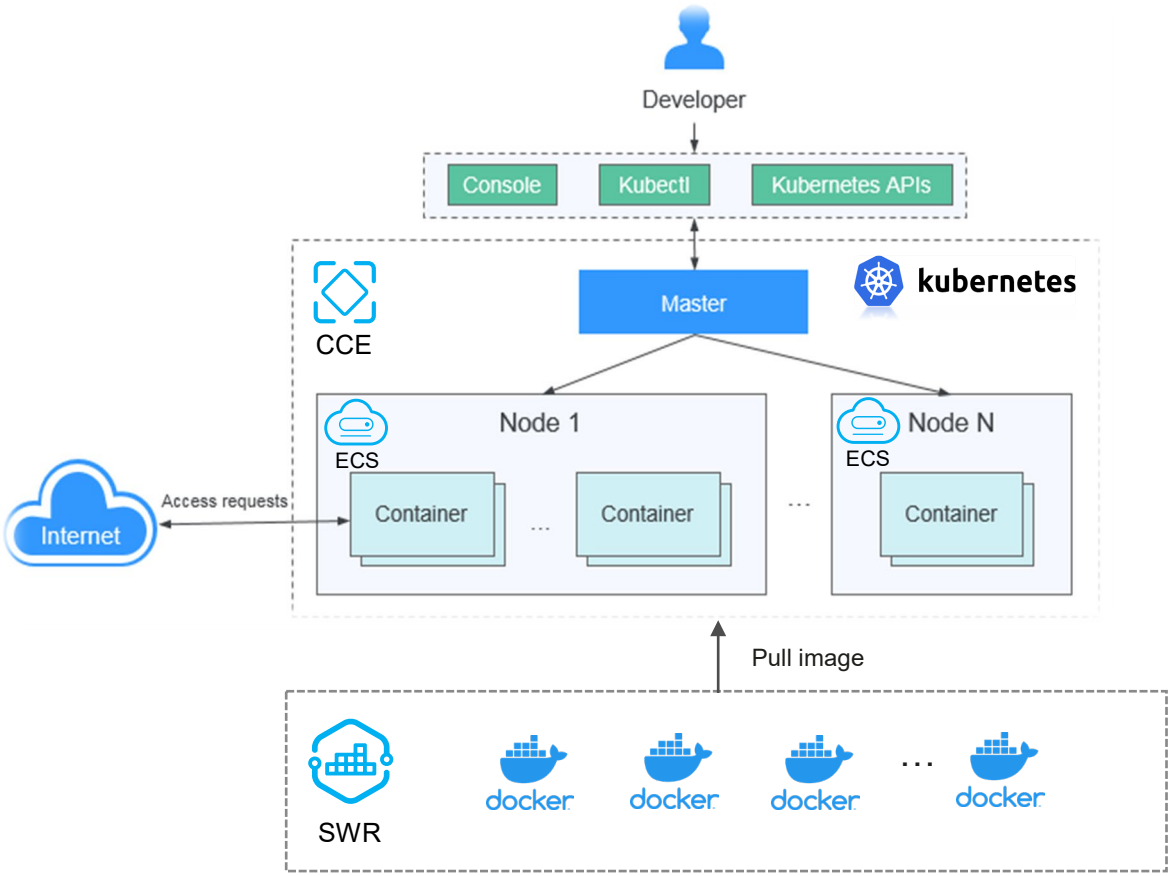
Container



CLOUD NATIVE COMPUTING FOUNDATION	docker	kubernetes	KubeEdge	Volcano
CNCF platinum member founding member **No.1** in Asia	Key member Contributed Commits **No.3**	**No.1** contributor in Asia **No.1** in terms of maintainer seats in Asia	**Initiator** of the container-based edge computing project	**Initiator** of the container-based batch computing project

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Cloud Container Engine (CCE)

One-stop deployment and operation and maintenance

One-click creation of Kubernetes clusters, automated deployment, operation and maintenance of container applications, and the entire life cycle is completed in a one-stop container service.

Deep integration of IaaS resources

Direct use of Huawei Cloud **ECS** / **BMS**, network (**VPC** / **EIP**), storage (**EVS** / **OBS** / **SFS**) capabilities. Compatible with native Kubernetes and Docker

High availability business without interruption

The cluster **control plane HA and cross-AZ** are highly available, container applications are elegantly scaled, and they go offline safely to ensure that services do not drop.

Software Repository for Container (SWR)

SWR allows you to securely host and efficiently distribute images on the cloud without building or maintaining image repositories by yourselves.

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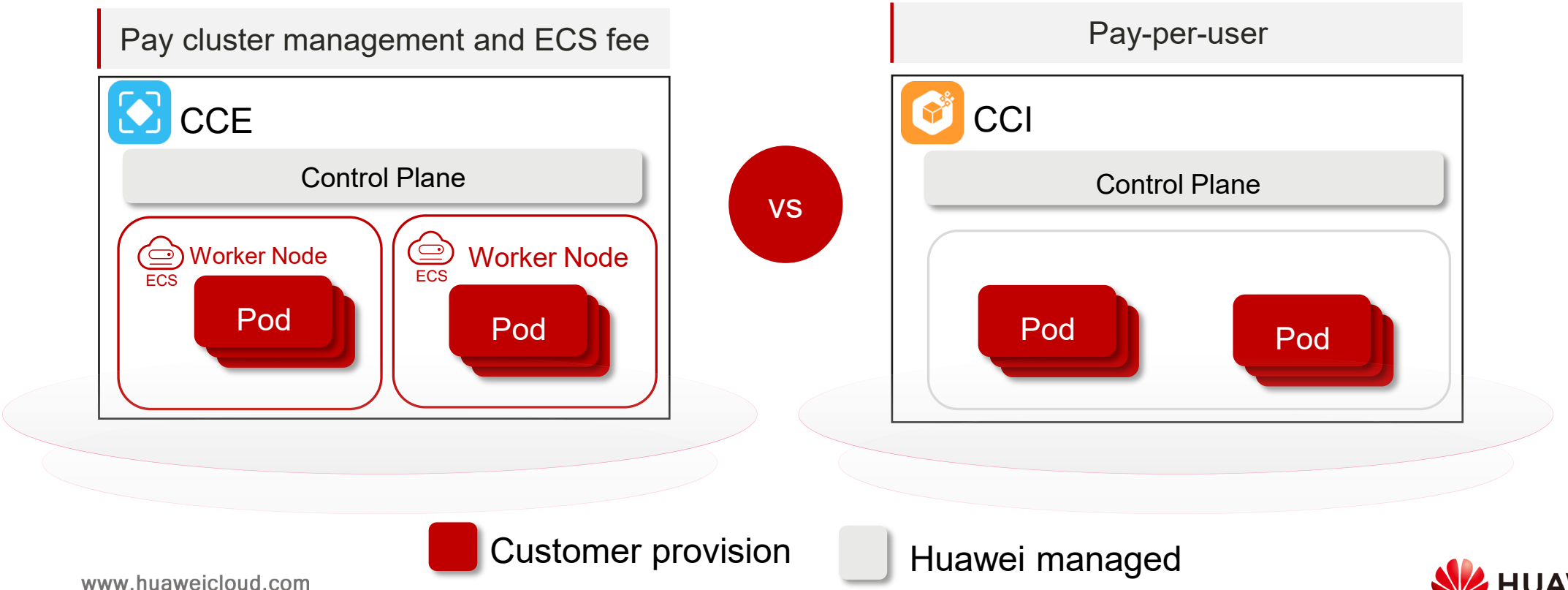
Container

Cloud Container Instance (CCI) is a **serverless container** engine that allows you to run containers without creating or managing server clusters. In the serverless model, a cloud provider runs servers and dynamically allocates resources so that you can build and run applications without having to worry about server statuses.

CCI pods adapt to customer requirements. Node management and capacity planning are not required, further reducing O&M costs.



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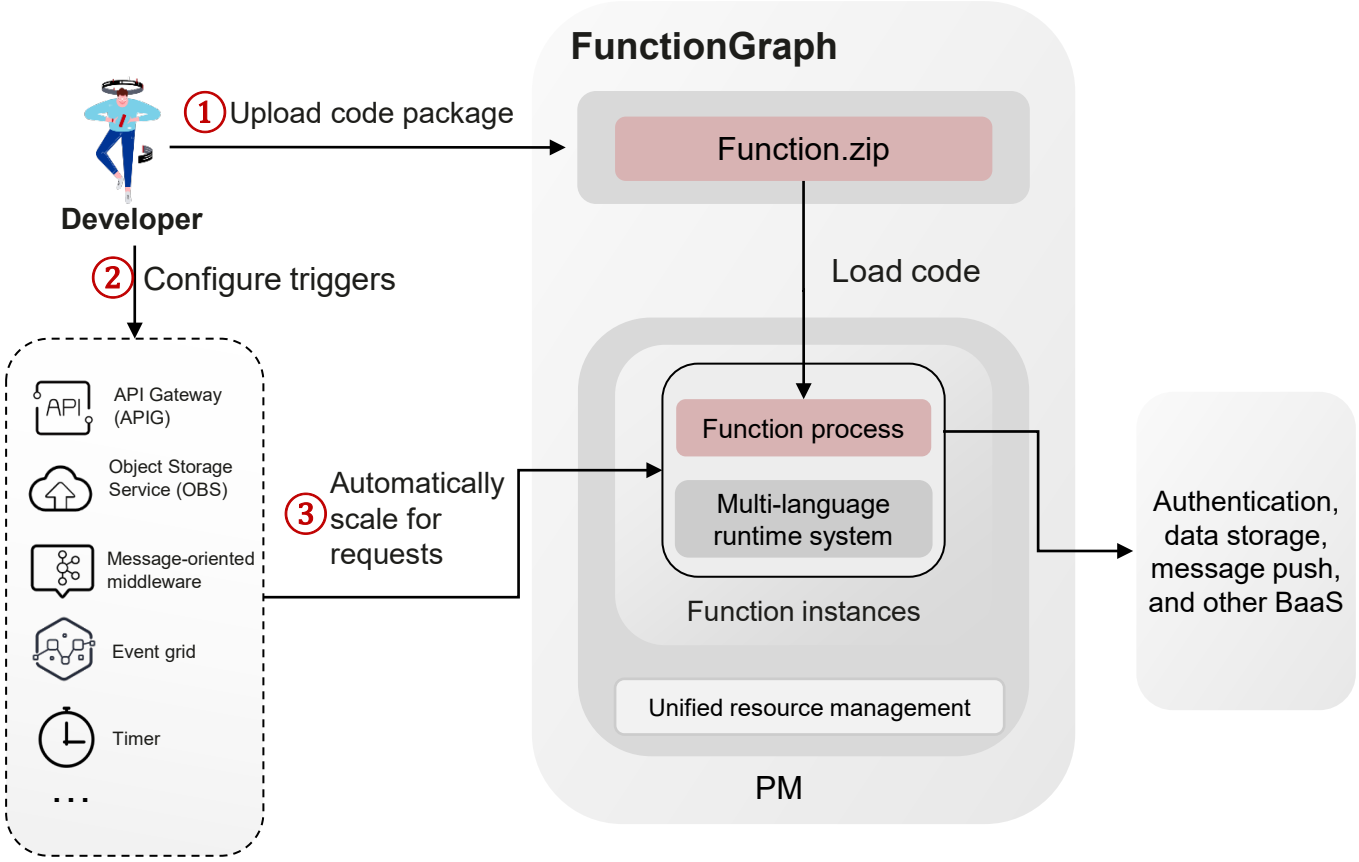
Function Graph

Function Graph allows you to **run your code without provisioning or managing servers**, while ensuring high availability and scalability. All you need to do is upload your code and set execution conditions, and Function Graph will take care of the rest . You pay only for what you use and you **are not charged when your code is not running.**



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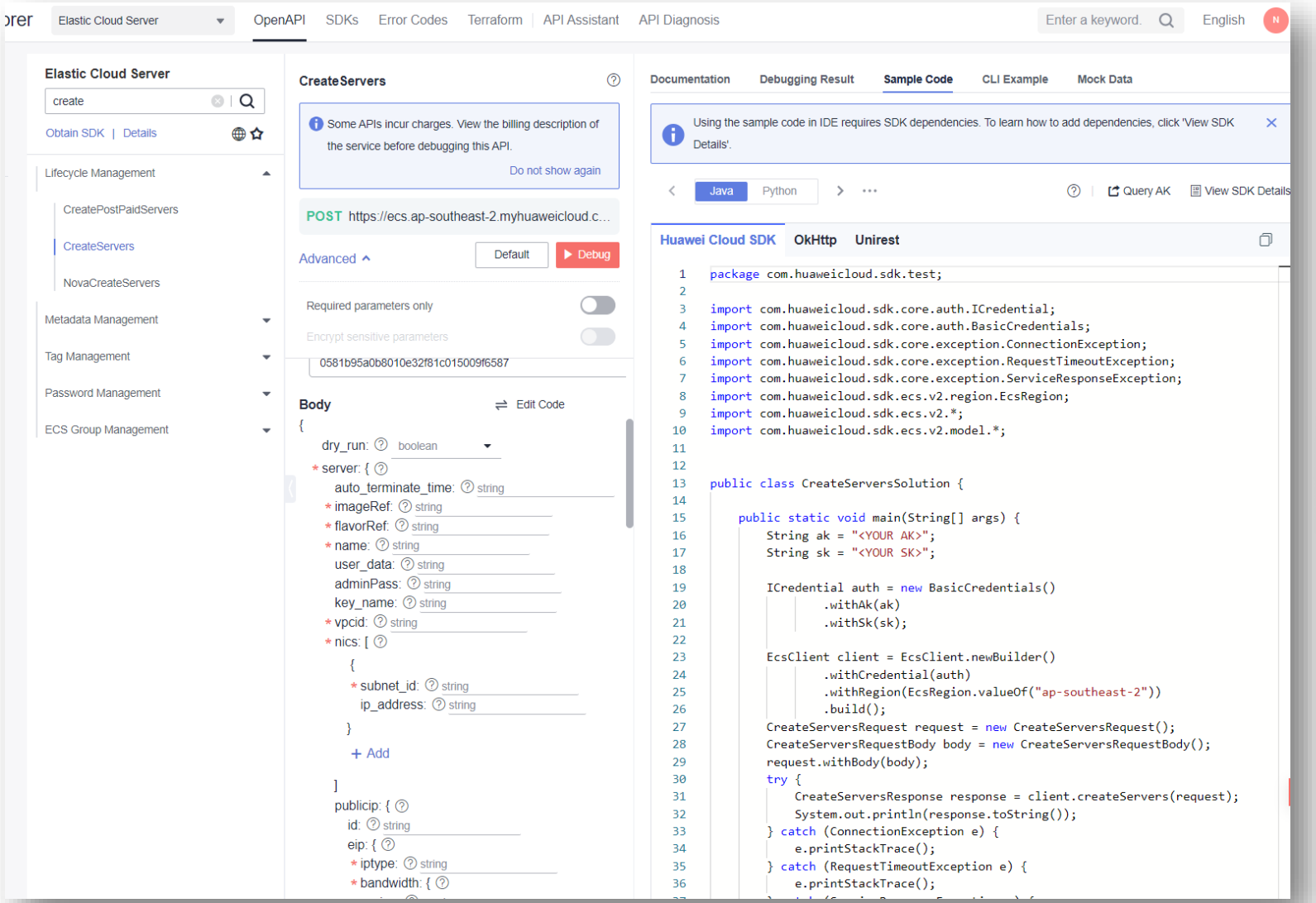
API Explorer



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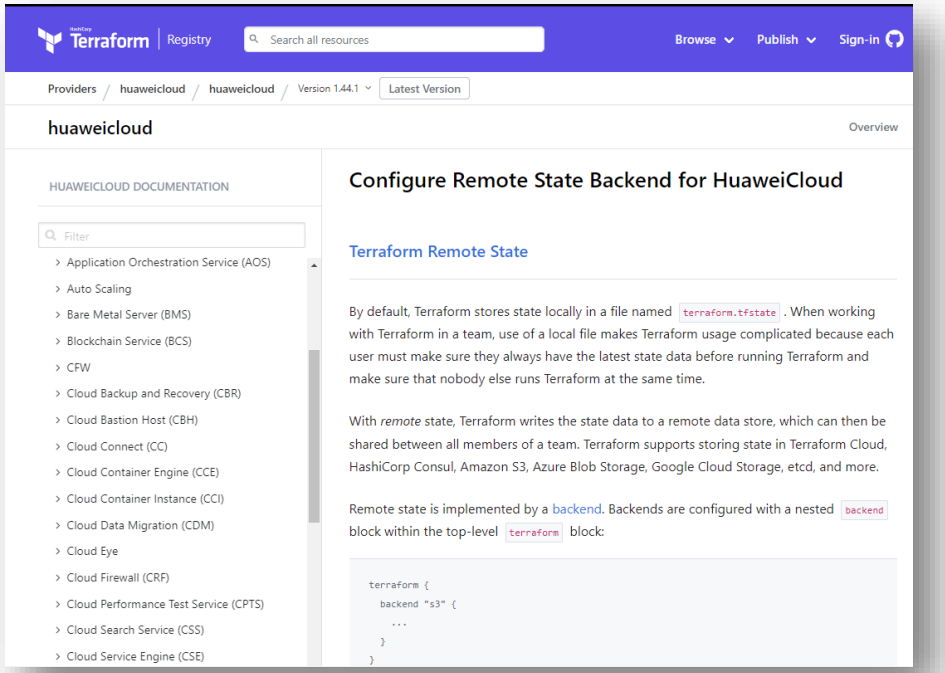
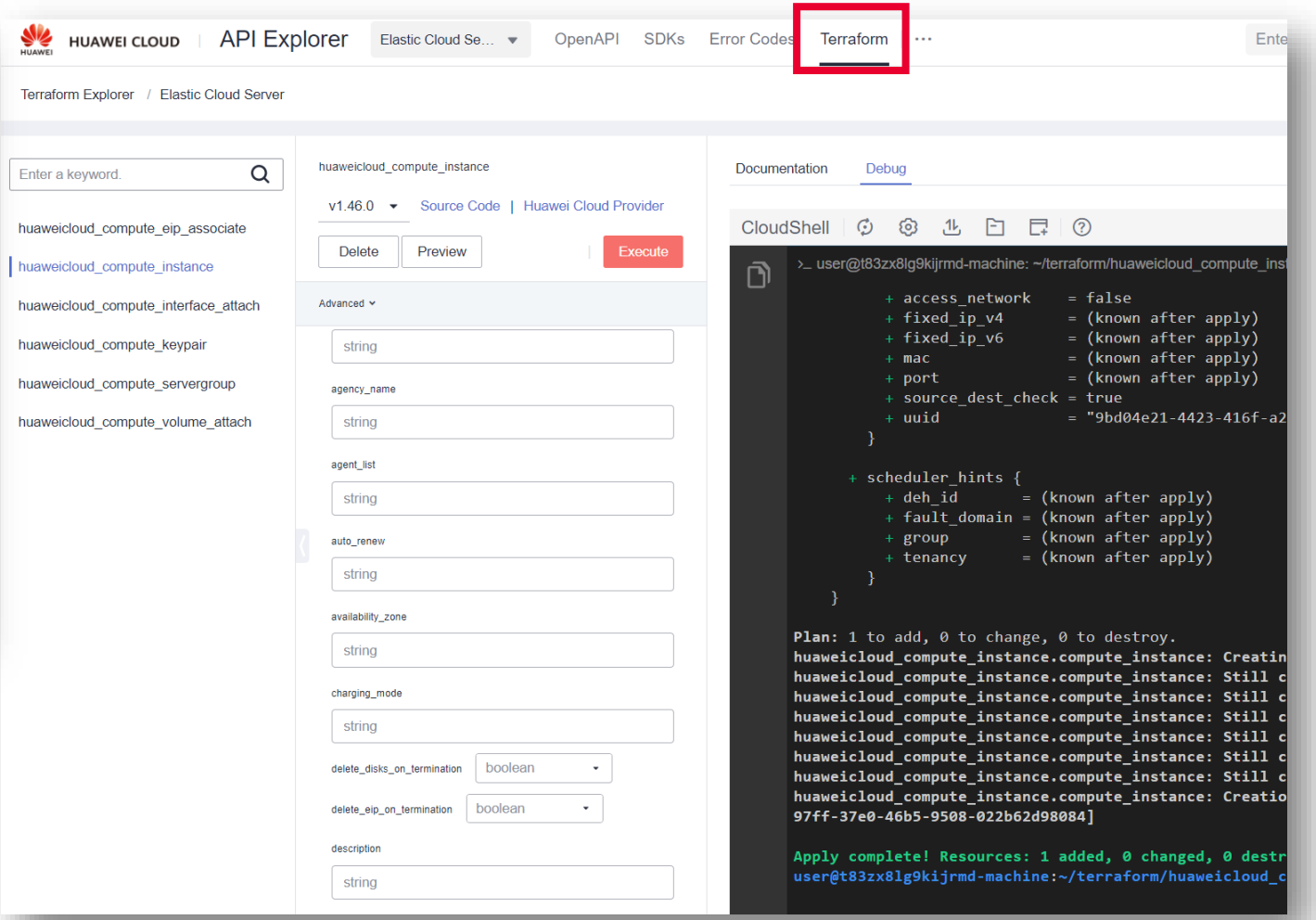
Terraform





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